

Spike Gate Samples Explained

Narrative guide to the bar-major profile spike gate used by `bar_spike_gated_foreground_report.py`.

One-sentence summary: the spike gate marks narrow, local, off-centre brightness peaks in the bar-major profile, then keeps only foreground-candidate mask segments that intersect those marked profile samples.

Why the spike gate exists

Foreground stars and compact artefacts often appear as narrow positive spikes when the galaxy image is sampled along the bar major axis. Real galaxy structure can also be bright, but it usually changes more gradually. The spike gate tries to separate those cases before deciding which residual-image segments should be used as a foreground mask.

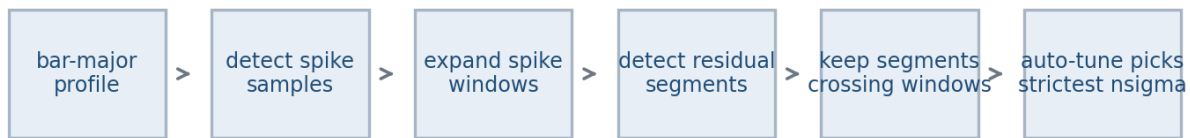


Figure 1. Overall spike-gated masking flow.

What is a spike sample?

The code inspects one profile sample at a time. A sample can become a spike only if it passes several tests: it must be finite and positive, it must not lie inside the central exclusion zone, it must be a local peak compared with its immediate neighbours, it must exceed a wider surrounding baseline, and it must drop away sharply over a short side offset.

- Candidate sample: the profile value currently being tested.
- Neighbour baseline: the median of profile samples a specified radial distance away from the candidate.
- Side samples: two nearby samples, one on each side of the candidate, used to test whether the peak is narrow.
- Expanded spike window: extra profile samples added around a detected spike so mask/profile intersections are not too brittle.

The side-drop test

The side-drop test is the most literal sharpness test. If the candidate is at index i , the code looks `spike_side_offset_samples` positions to the left and right. It takes the median of those two side values and requires the candidate to be brighter by `spike_side_drop_fraction`.

In formula form: $\text{candidate} \geq (1 + \text{spike_side_drop_fraction}) \times \text{median}(\text{left side sample}, \text{right side sample})$.

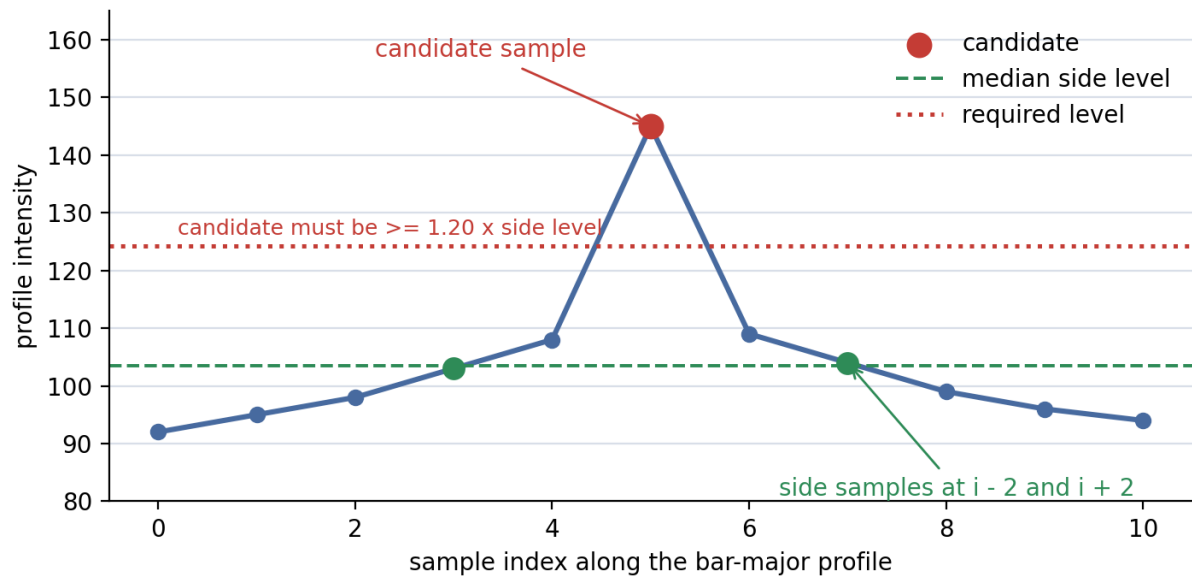


Figure 2. A candidate sample passing the side-drop test with `offset = 2` and `side-drop fraction = 0.20`.

Quantity	Example value	Meaning
left side sample	108	Profile value at $i - 2$.
right side sample	109	Profile value at $i + 2$.
side level	108.5	Median of the two side samples.
side-drop fraction	0.20	Candidate must be 20% above the side level.
required candidate value	130.2	108.5×1.20 .
actual candidate value	145	Passes, because $145 \geq 130.2$.

Interpretation: increasing `spike_side_drop_fraction` demands a pointier peak; decreasing it allows broader, gentler peaks to pass.

The wider neighbour baseline

The side-drop test asks whether the peak is sharp close to the candidate. The wider neighbour-baseline test asks whether the candidate is high compared with its surrounding profile context. The code gathers samples whose

distance from the candidate lies between `spike_neighbour_inner_arcsec` and `spike_neighbour_outer_arcsec`. It then requires the candidate to exceed the median of those neighbour samples by `spike_excess_fraction`.

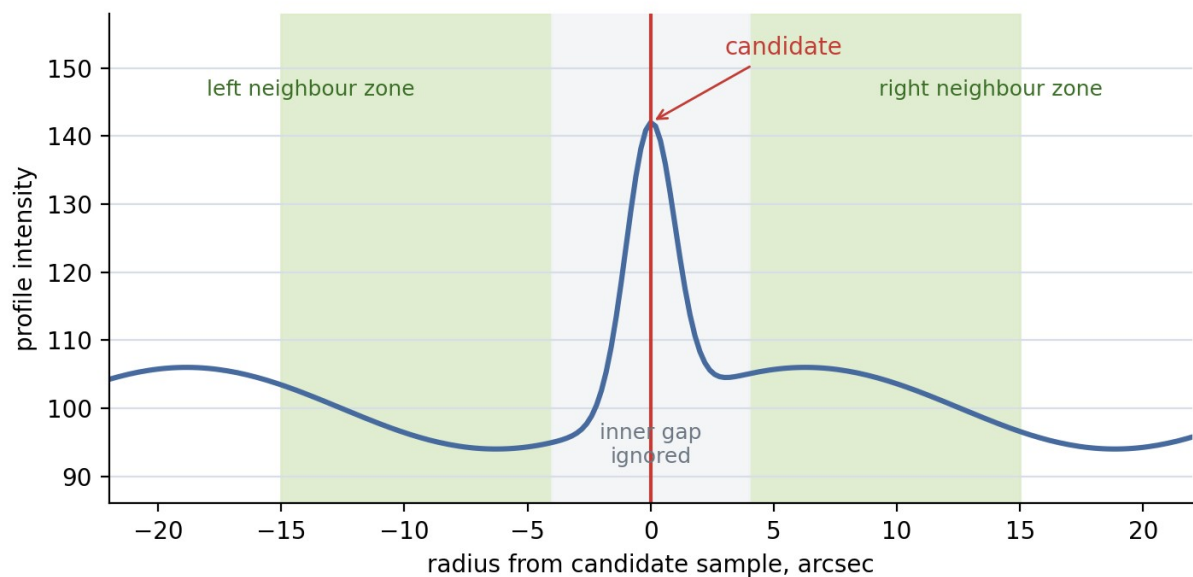


Figure 3. The inner gap is ignored; neighbour samples come from the green zones.

Quantity	Example value	Meaning
<code>spike_neighbour_inner_arcsec</code>	4 arcsec	Samples closer than this are not part of the baseline.
<code>spike_neighbour_outer_arcsec</code>	15 arcsec	Samples farther than this are not part of the baseline.
<code>neighbour median</code>	100	Typical surrounding profile level.
<code>spike_excess_fraction</code>	0.25	Candidate must be 25% above the neighbour median.
<code>required candidate value</code>	125	100×1.25 .
<code>actual candidate value</code>	145	Passes, because $145 \geq 125$.

Why both tests? the neighbour baseline rejects peaks that are not high relative to their wider surroundings, while the side-drop test rejects broad galaxy humps that do not fall away quickly near the candidate.

The central exclusion

The galaxy centre is often bright, structured, and steep. A central peak is therefore not good evidence for a foreground contaminant. The parameter `spike_center_exclusion_arcsec` defines a symmetric interval around zero radius where candidates are skipped before the spike tests are applied.

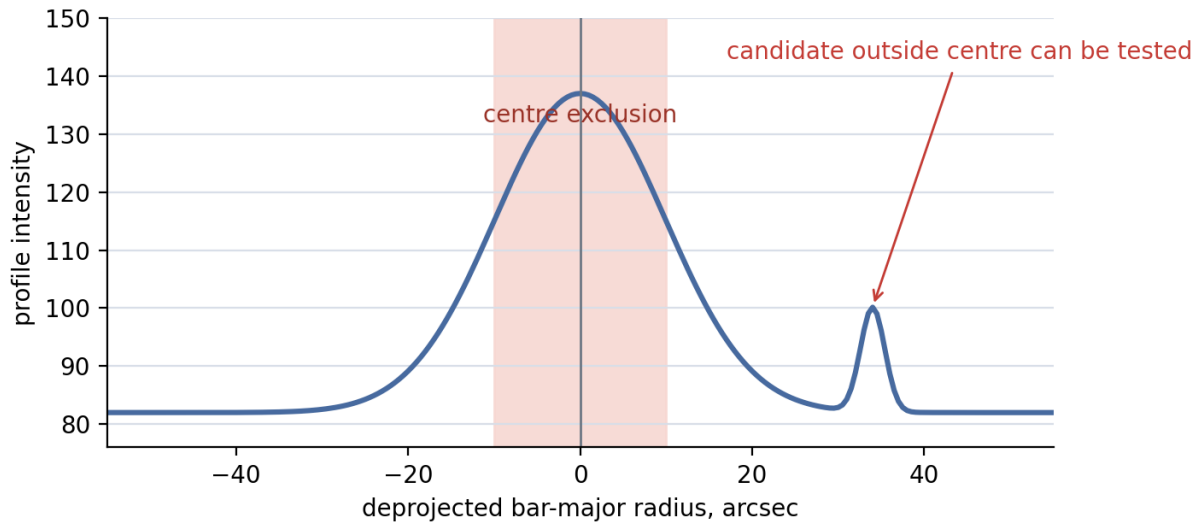


Figure 4. Candidate samples inside the centre exclusion are ignored.

Sample radius	centre exclusion	Tested?
-6 arcsec	10 arcsec	No. $\text{abs}(\text{radius}) < 10$.
0 arcsec	10 arcsec	No. This is the galaxy centre.
+8 arcsec	10 arcsec	No. $\text{abs}(\text{radius}) < 10$.
+18 arcsec	10 arcsec	Yes. It is outside the excluded centre.

Spike-window expansion and coverage

After a sample passes the spike tests, the code expands it along the one-dimensional profile by `spike_window_samples`. This does not create more detections in the image. It simply widens the bar-profile gate used to decide whether a residual-image segment intersects the spike.

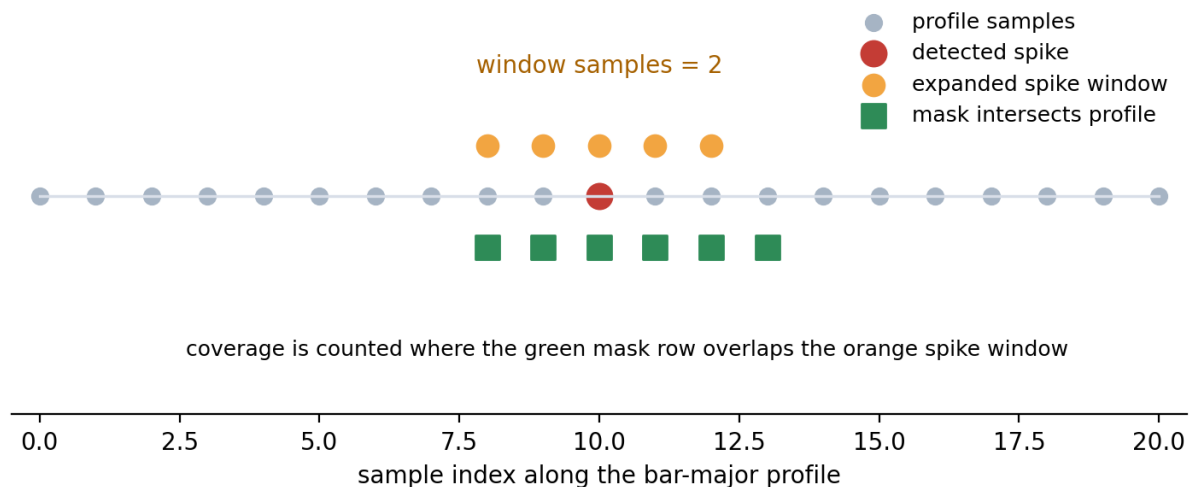


Figure 5. A detected spike sample expanded by two samples on each side.

Parameter	Example	Result
detected spike index	10	The profile sample that passed the spike tests.
spike_window_samples	2	Expand to samples 8, 9, 10, 11, and 12.
mask/profile intersection	samples 8 to 13	The mask covers all expanded spike samples.
coverage	$5 / 5 = 1.0$	Auto-tune treats the spike samples as fully covered.

How this affects auto-tune nsigma

In spike-gated mode, auto-tune does not invent a new threshold from first principles. It tries the candidate thresholds in `auto_tune_nsigmas` from conservative to aggressive, usually 5.0, 4.5, 4.0, and 3.5. For each threshold it detects compact residual-image segments, filters them, keeps only segments that intersect the expanded spike samples, and measures coverage.

The selected nsigma is the first, most conservative threshold that covers essentially every detected spike sample. If the spike gate detects weaker or more numerous spikes, auto-tune may need a lower, more aggressive nsigma. If the spike gate detects only obvious sharp spikes, a higher, more conservative nsigma may be enough.

Candidate nsigma	Mask segments found?	Spike-window coverage	Auto-tune decision
5.0	1 segment	60%	Not enough coverage. Try next threshold.
4.5	2 segments	100%	Select 4.5, because it is the first full-coverage threshold.
4.0	More segments	100%	Not needed; 4.5 already worked.
3.5	Most segments	100%	Fallback only if stricter candidates fail.

Parameter cheat sheet

Parameter	Controls	If increased
<code>spike_excess_fraction</code>	How high the candidate must be above the wider neighbour baseline.	Fewer, stronger spikes pass.
<code>spike_neighbour_inner_arcsec</code>	How close to the candidate the baseline zone begins.	Baseline ignores more near-candidate structure.
<code>spike_neighbour_outer_arcsec</code>	How far from the candidate the baseline zone extends.	Baseline uses a wider radial context.

Parameter	Controls	If increased
spike_side_offset_samples	How far left/right the sharpness side samples are.	Sharpness is tested over a broader local span.
spike_side_drop_fraction	How much the candidate must exceed those side samples.	Only pointier peaks pass.
spike_center_exclusion_arcsec	Radius around the galaxy centre where spike testing is skipped.	More central structure is ignored.
spike_window_samples	How many neighbouring profile samples are added around each detected spike.	Coverage becomes less brittle but more permissive.

Practical reading of the gate

A detected spike sample is therefore not just a bright point. It is a bright point that is off-centre, locally peak-like, high relative to a wider radial neighbourhood, and sharp relative to nearby side samples. The expanded spike window then turns that point decision into a small target region. The final foreground mask keeps residual-image segments only when they cross those target regions.

Systematic documentation review - 17 August 2026

Review classification: Reviewed - explanatory reference remains valid

Updates made: Definitions of spike samples, side-drop tests, neighbour baselines and central exclusion were reviewed; no Toy Objects-specific correction was required.

Current interpretation: Use this document to interpret Spike Gate samples only. It is independent of the final Toy Objects method comparison.

Authoritative combined outcome: SEP and MTOObjects Masking Comparison and Recommendations 20260817.docx.
Full optimisation history and final science-image SEP fold records: Foreground Masking Optimisation Results.xlsx.